

Implementation Request — Bridge Layer

Identity Entropy Output Layer · Formation Context Wiring · Coalescence Tracking

Doc ID	CR-2026-01	Status	Open — pending implementation
Target	v5.2.0-internal	Audience	VSEPR-SIM / VSIM developers
Raised	2026-05-31	Closed	—

Ref: Theoretical-Simulation Bridge Project (`vsim_bridge_doc.tex`) — read that first. This document assumes familiarity with § 1–10 of the bridge doc and does not re-derive theory.

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§1 Summary

The bridge doc defines what the identity entropy layer should do. This document converts that into ten concrete implementation tasks with priorities, interface contracts, open decisions, and acceptance criteria. Tasks are grouped into three priority tiers. **P0** tasks are blocking for v5.2.0. **P1** tasks complete the feature. **P2** tasks can defer to a follow-on milestone.

Total tasks	P0 — blocking	P1 — required	P2 — deferrable	Open decisions
10	3	5	2	5

§2 Implementation Tasks

Dependencies: CR-03 and CR-04 must land before CR-05 through CR-09 can be meaningfully tested. CR-01 must land before CR-07. CR-06 has a hard dependency on DEC-05 being resolved.

ID	Task	Ref	Pri.	Status
CR-01	Parse [formation] and [formation.initial_state] into FormationContext. Struct skeleton in vsim/identity_entropy.hpp. Fields: method, history, formation_time, temperature_K, pressure_Pa, phase, defect_state, grain_state, density. Absent sub-fields default to "unknown" / zero, not error.	§4, §10	P0	PENDING
CR-02	Formation validation pass — mark runs with absent or incomplete formation as incomplete in the validation layer. Applies to simulations with material types: solid, surface, gas, pipe, powder, structural. Missing [formation] block produces a validation warning, not a hard failure. Runs must still execute.	§4	P0	PENDING
CR-03	Parse [outputs.identity_entropy] config block and wire into the export pipeline. Fields: enabled, path, format, precision, attach_to_viewer, write_per_frame, write_summary, variables. If enabled = false the block is parsed but no writer is instantiated. Schema decision DEC-01 must be resolved before this lands.	§5	P0	PENDING

ID	Task	Ref	Pri.	Status
CR-04	Implement IEWriter — header, per-frame write, summary, close. Full implementation stub in <code>vsim/identity_entropy.cpp</code> (bridge doc §10). <code>write(IERecord&)</code> must call <code>sync()</code> before writing. Writer must not throw after <code>close()</code> is called. Buffer strategy to be confirmed per DEC-04.	§5, §10	P0	PENDING
CR-05	Populate all 12 IERecord fields per simulation frame. <code>J</code> , <code>data_loss</code> , <code>recoverability</code> have defined derivation rules. <code>data_entropy</code> , <code>hidden_activity</code> , and <code>projection_loss</code> require physics model input — see DEC-02. <code>state_quota</code> maps to Q_n . <code>entropy_flux</code> is the frame-delta of identity entropy. Fields not yet computable emit 0.0 with flag bit set.	§6	P1	PENDING
CR-06	Wire coalescence_gate() into the bonding/clustering event system. When the gate fires, emit nonzero <code>coalescence_loss</code> and <code>residual</code> into the <code>IERecord</code> for that frame. Gate is <i>read-only</i> — it must not mutate particle state on fire. <code>coalescence_residual()</code> provides the scalar loss value. See DEC-05.	§7, §10	P1	PENDING
CR-07	Implement <code>FormationContext::formation_memory()</code> and wire its result into <code>IERecord.formation_memory</code> each frame. The heuristic in bridge doc §10 scores six declared descriptors. A real implementation should degrade M_f over time as the system thermalizes. The heuristic is the starting floor. Thermalization model is out of scope for P1; static heuristic is sufficient to unblock.	§8, §10	P1	PENDING
CR-08	Add "identity_entropy" as a valid entry in <code>[observe].track</code>. When present, the runtime activates identity entropy tracking. If the entry is present but <code>[outputs.identity_entropy]</code> is absent or <code>enabled = false</code> , emit a validation warning (not a failure).	§9	P1	PENDING
CR-09	Per-frame vs summary-only write modes, controlled by <code>write_per_frame</code> and <code>write_summary</code> flags. <code>write_per_frame = false + write_summary = true</code> produces a summary-only .ie file. Both false is a no-op that still creates the file with just the header. Both true is the default full output.	§5	P1	PENDING

CR-10	Viewer overlay wiring when <code>attach_to_viewer = true</code>. The <code>.ie</code> file path should be registered with the viewer's overlay system. Exact overlay API is viewer-team's call — this task is just the registration handoff from the export pipeline. Can ship without this; viewer overlay is not required for the output layer to be useful.	§5	P2	PENDING
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§3 Interface Contracts

These constraints must not be violated regardless of how tasks are subdivided internally. They follow directly from the bridge doc's core principle.

IC-01	Identity entropy variables (<code>J</code> , <code>data_loss</code> , all <code>IERecord</code> fields) must never be written back into mutable simulation state. They are derived output only.
IC-02	The <code>coalescence_gate()</code> function is read-only. A gate fire produces a sidecar record. It does not create, remove, or modify particles.
IC-03	Render state must not mutate truth state. The viewer overlay (CR-10) reads from the <code>.ie</code> file; it does not feed back into the simulation kernel.
IC-04	<code>FormationContext</code> is a runtime-only object. It must not be serialised into <code>.dynx</code> or any ground-truth particle record.
IC-05	<code>IEWriter::write(IERecord&)</code> must call <code>sync()</code> before writing. Callers must not pre-call <code>sync()</code> and assume the writer will skip it — double-sync is safe; missing sync is not.
IC-06	A missing or disabled <code>[outputs.identity_entropy]</code> block must not degrade simulation performance. The code path must be a no-op when disabled.
IC-07	<code>J</code> must be in the range <code>[0.0, 1.0]</code> at all times. Clamp on computation, not on write. Values outside range indicate a model error.

§4 Open Decisions

These must be resolved before the tasks that depend on them can land.

DEC-01	Schema bracket: <code>[outputs.identity_entropy]</code> or <code>[export.identity_entropy]</code> ? The bridge doc deliberately deferred this. Pick one and stick to it. Blocks CR-03. If the rest of the export system uses <code>[export.*]</code> sub-tables, consistency suggests folding in. If <code>[outputs.*]</code> is the new direction, keep it separate.	TBD
DEC-02	How are <code>dataentropy</code> , <code>hidden_activity</code> , and <code>projection_loss</code> computed? These require a model of the projection operator from identity-space to observable state. The bridge doc defines the concept but not the formula. Until resolved, CR-05 emits <code>0.0</code> with flag bit <code>0x1</code> set. Blocks CR-05 (full), CR-06 (partial).	TBD
DEC-03	<code>lineage_depth</code> tracking strategy: per-object chain or global lineage registry?	TBD

Each object could carry its own depth counter (simple, cheap, no global state), or the runtime could maintain a lineage map keyed on identity ID (accurate for re-coalesced objects, more memory). Blocks CR-05 for `lineage_depth` population.

DEC-04**TBD**

`IEWriter` buffer strategy: stream per frame or accumulate N frames then flush? Streaming per frame is simpler and crash-safe. Buffering N frames is faster for large simulations. The summary pass already buffers all records in memory — if memory is a concern the summary could be a running accumulator. Blocks CR-04 finalisation.

DEC-05**TBD**

Definition of the projection operator A_n for coalescence residual computation. `coalescence_residual()` takes caller-supplied `projected_identity` vector. Who computes the projection, and how? The bridge doc defines $R_{\text{coal}}^{(n)} = C_n - A_n(I_{(n+1)}^*)$ but A_n is not yet specified. Blocks CR-06.

§ 5 Acceptance Criteria

A milestone is shippable when all **P0** and **P1** criteria below are checked off. **P2** criteria are tracked but not required for v5.2.0.

ID	Tick	Criterion	Priority
AC-01	☐	A surface deposition run with <code>[outputs.identity_entropy] enabled = true</code> produces a valid <code>.ie</code> file at the declared path.	P0
AC-02	☐	The <code>.ie</code> file contains a header row, at least one data row per frame, and a <code># SUMMARY</code> block with <code>rows</code> , <code>mean_J</code> , <code>min_J</code> , and <code>coal_events</code> .	P0
AC-03	☐	J is within $[0.0, 1.0]$ for all rows. <code>data_loss</code> equals $1 - J$ to within float64 tolerance. <code>recoverability</code> equals $1 - \text{projection_loss}$.	P1
AC-04	☐	A run with all six formation descriptors declared produces <code>formation_memory = 1.0</code> on frame 0. A run with no formation block produces <code>formation_memory = 0.0</code> .	P1
AC-05	☐	A run without a <code>[formation]</code> block executes successfully and the validation report contains a <code>formation: incomplete</code> warning. No hard failure is emitted.	P0
AC-06	☐	When a coalescence event fires, the corresponding frame row in the <code>.ie</code> file has <code>coalescence_loss > 0.0</code> . Frames without coalescence events have <code>coalescence_loss = 0.0</code> .	P1
AC-07	☐	<code>[outputs.identity_entropy] enabled = false</code> produces no <code>.ie</code> file and no measurable performance overhead versus a run without the block.	P0
AC-08	☐	A script with <code>"identity_entropy"</code> in <code>[observe].track</code> but no <code>[outputs.identity_entropy]</code> block emits a validation warning and runs without error.	P1
AC-09	☐	<code>write_per_frame = false + write_summary = true</code> produces a <code>.ie</code> file with only the header and <code># SUMMARY</code> block — no data rows.	P1

- AC-10 ☐ With `attach_to_viewer = true`, the `.ie` file path is registered with the viewer overlay system. The viewer can display per-identity entropy values without the simulation re-running. P2

§6 File Inventory

New files introduced by this request. Existing files should not require structural changes beyond adding new parser branches and wiring calls.

File	Type	Contents	Tasks
<code>vsim/identity_entropy.h</code>	New header	FormationContext, IERecord, CoalescenceInput, <code>coalescence_gate()</code> , <code>coalescence_residual()</code> , IEWriter. Skeleton in bridge doc § 10.	CR-01, 04, 06, 07
<code>vsim/identity_entropy.c</code>	New TU	Implementations of all non-inline functions from the header. Skeleton in bridge doc § 10.	CR-04, 06, 07
<code>vsim/parser/formation_parser.*</code>	New parser module	Reads <code>[formation]</code> and <code>[formation.initial_state]</code> into FormationContext. Follows existing parser conventions.	CR-01
<code>vsim/export/ie_export.*</code>	New export module	Reads <code>[outputs.identity_entropy]</code> config, instantiates IEWriter, wires frame callbacks into existing export pipeline alongside <code>.dynx</code> , analysis JSON, etc.	CR-03, 05, 08, 09
<code>vsim/validation/formation_check.*</code>	New validation check	Emits formation: incomplete warning for runs missing <code>[formation]</code> . Runs after config parse, before runtime init.	CR-02

Path convention: Module paths above follow the existing VSIM source tree convention and are illustrative. Actual paths should follow whatever the team has established for v5.x. Do not bikeshed the filenames; do bikeshed the interfaces.